



Prashast Sidhant

RollNo.: 23113049

Bachelor of Technology

Chemical Engineering

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Education

Degree/Board	Institution	CGPA/%
B.Tech., Chemical Engineering	National Institute of Technology Raipur	8.11/10.0
CBSE - XII	Kendriya Vidyalaya Bailey Road, Patna, Bihar	92.2%
CBSE - X	Kendriya Vidyalaya Bailey Road, Patna, Bihar	94.2%

Experience

Space Applications Centre (SAC), ISRO

Research Intern - Generative AI | Ahmedabad

May 2026 - Jul 2026

- Selected for the internship through the Bharatiya Antariksh Hackathon 2025 to develop an AI-based information retrieval solution for the MOSDAC portal.
- Collaborated with SAC scientists to translate information-retrieval requirements into a deployable offline Generative AI system.
- Built an offline Hybrid RAG system combining vector search, BM25 retrieval and a Neo4j Knowledge Graph for contextual retrieval from official scientific documents.
- Created Python workflows for Web Scraping, document extraction, validation, chunking, embedding generation and entity-relationship extraction.
- Developed FastAPI-based REST services and integrated LLM, ChromaDB, Neo4j and Docker with citation verification and AI guardrails.

Personal Projects

- MedTwin: Autonomous GenAI Digital Twin Framework for Medical Devices** (National Finalist - Tata Elxsi Teliport Season 3)
 - Developed an AI Digital Twin for automated and compliant system design of Class II/III medical devices.
 - Built using Python, LangGraph, FastAPI, React, WebSockets, PostgreSQL, Redis and LLMs; accelerated development using parallel multi-agent workflows and multi-physics simulation.
- Smriti-Saarthi - Founder | AI-Powered Adaptive Learning Platform**
 - Selected for Pre-Incubation at NITRRFIE under the NIDHI Technology Business Incubator.
 - Building an autonomous AI mentor for competitive-exam aspirants using Agentic RAG, knowledge graphs, spaced repetition and adaptive question generation.
 - Designed with Python, FastAPI, React, Azure, LLMs, vector search and multilingual voice assistance.
- Online Payment Fraud Detection using Machine Learning** (Top 11 Project - CEC Certification Course on Applied Data Analytics)
 - Developed a machine-learning model to classify online transactions as fraudulent or legitimate.
 - Performed data preprocessing, statistical analysis, model training and visualization using Python, Pandas, NumPy, Scikit-learn and Matplotlib.

Technical Skills and Interests

Languages & Frameworks: Python, C++, JavaScript, SQL, FastAPI, REST APIs, LangChain, LangGraph, CrewAI, Streamlit.

AI & Data: LLMs, RAG, Neo4j Knowledge Graph, vector databases, Scikit-learn, Pandas, NumPy, machine learning and prompt engineering.

Developer Tools: Git & GitHub, PostgreSQL, Azure Services, Jupyter Notebook, Visual Studio, Cursor IDE and Gemini CLI.

Soft Skills: English communication, public speaking, decision-making, adaptability, teamwork and leadership.

Positions of Responsibility

- Selected (Team Leader) in two consecutive editions for the Bharatiya Antariksh Hackathon**, 2025 and 2026.
- Core Coordinator, Team Technocracy, NIT Raipur** - jury member for Case Ops 4.0 and organizer of the national-level Dev Clash hackathon.
- Group Leader** of a top 11 project in the CEC Certification Course on Applied Data Analytics.
- Team Leader** - National Finalist, NABARD Agri Credit Hackathon; National Finalist, Tata Elxsi Teliport Season 3.
- Space Tech Entrepreneur Intern** at India Space Lab (virtual); nominated team leader for Smart India Hackathon 2024.
- Startup Idea Selected for Pre-Incubation at the NIT Raipur Foundation for Innovation & Entrepreneurship (NITRRFIE)** under the NIDHI Technology Business Incubator for the venture's innovation and potential impact.

Achievements

- Secured 96.2 percentile in JEE Main 2023.
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional.
- Completed McKinsey & Company's Forward Program in leadership and problem solving.
- Ranked 818 among 7,000+ submissions in the Amazon ML Challenge.
- Recognized among the Top 80 Unstop Competitive E-School Leaders at the Unstop Talent Awards 2026.
- National Semi-Finalist - Novartis NEST 2.0; detailed presentation round - HP Power Lab 2.0.